

CLASSROOM CONTACT PROGRAMME

(ACADEMIC SESSION 2019-2020)

PRE-MEDICAL : ENTHUSIAST COURSE [PHASE-II]

TEST # 06

DATE : 04-08-2019

Test Type : Minor

Test Pattern : AIIMS

INSTRUCTIONS

Do not open this Test Booklet until you are asked to do so

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The test is of **3 hours** duration.
4. The Test Booklet consists of **180** questions. The maximum marks are **180**.
5. **Each correct answer carry 1 mark. While one third mark will be deducted for each incorrect answer.**
6. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** the Answer Sheet. **Use of pencil is strictly prohibited.**
7. No candidate is allowed to carry any textual material, printed or written, bits of papers, pager, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
8. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
9. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
10. **Do not fold or make any stray marks on the Answer Sheet.**

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MINOR TEST # 06

Topics

DATE : 04 - 08 - 2019

PRE-MEDICAL ENTHUSIAST COURSE : PHASE - II – PATTERN : [AIIMS]

PHYSICS : Electrostatics, Current Electricity, Capacitance, (Till 1st August)

CHEMISTRY : GOC (Complete)

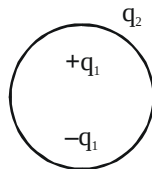
BIOLOGY : Mendalian Genetics, Molecular Genetcs (Upto DNA Replication), Human
Reproduction (Complete)

SPACE FOR ROUGH WORK

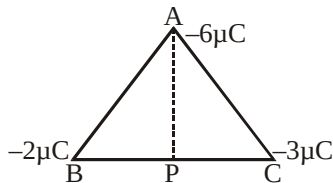
HAVE CONTROL → HAVE PATIENCE → HAVE CONFIDENCE ⇒ 100% SUCCESS

BEWARE OF NEGATIVE MARKING

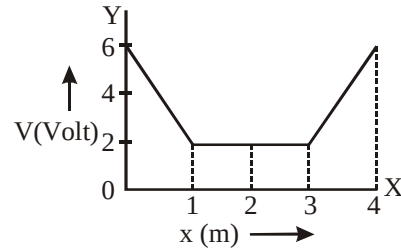
1. Consider the charge configuration and a spherical Gaussian surface as shown in the figure. When calculating the flux of the electric field over the spherical surface, the electric field will be due to:-



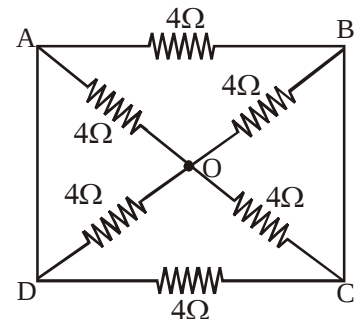
- (1) q_2
(2) only the positive charges
(3) all the charges
(4) $+q_1$ and $-q_1$
2. ABC is equilateral triangle of side 1m. Charges are placed at its corners as shown in fig. P is the mid-point of side BC, the potential at point (P) is-



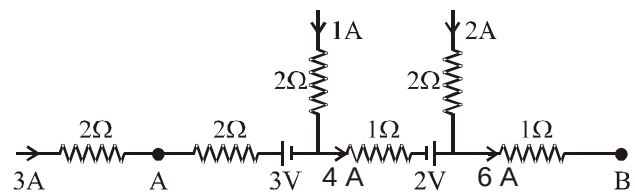
- (1) 2.7×10^3 V
(2) 1.52×10^5 V
(3) 1.3×10^3 V
(4) -1.52×10^5 V
3. The variation of electric potential with distance from a fixed point is shown in figure. What is the value of electric field at $x = 0.5$ m (in N/C) is :



- (1) Zero (2) 3 (3) 4 (4) 2
4. Six equal resistances each of $4\ \Omega$ are connected to form a Fig. The resistance between two corners A and B is :



- (1) $4\ \Omega$ (2) $4/3\ \Omega$
(3) $12\ \Omega$ (4) $2\ \Omega$
5. The potential difference between points A and B, in a section of a circuit shown, is :-



- (1) 5 volt (2) 1 volt
(3) 10 volt (4) 17 volt

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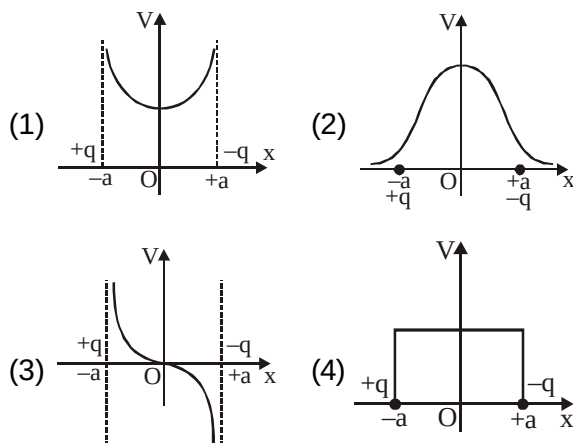
6. Two isolated charged conducting spheres of radii a and b produces the same electric field on their surfaces, the ratio of electric potentials on their surfaces is :-

(1) $\frac{a}{b}$ (2) $\frac{b}{a}$ (3) $\frac{a^2}{b^2}$ (4) $\frac{b^2}{a^2}$

7. The volume of three wires of copper are equal and their lengths are in the ratio $5 : 3 : 1$. The ratio of their electrical resistances is :-

(1) $1 : 9 : 25$ (2) $25 : 9 : 1$
(3) $1 : 15 : 125$ (4) $125 : 15 : 1$

8. A positive charge $(+q)$ and a negative charge $(-q)$ are placed at $x = -a$ and $x = +a$ respectively. Then the graph of V along y -axis and x along the x -axis is :-



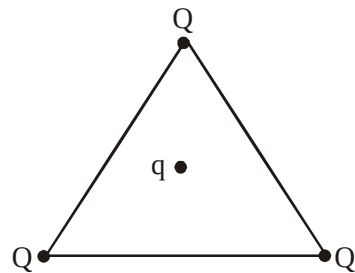
9. An electric charge q exerts a force F on a similar electric charge q separated by a distance r . A third charge $q/4$ is placed midway between the two charges. Now, the force F will :-

(1) become $F/3$ (2) become $F/9$
(3) become $F/27$ (4) remain F

10. A hollow metallic sphere of radius 8 cm is given a charge of 3.2×10^{-9} C. The electric potential at a distance 5 cm from its centre is :-

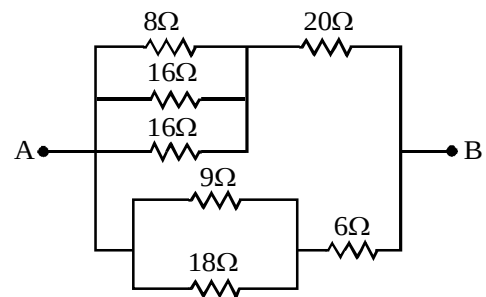
(1) 3.6×10^2 V (2) 9×10^2 V
(3) zero (4) 28.8×10^{-9} V

11. Three charges (each Q) are placed at the three corners of an equilateral triangle. A fourth charge q is placed at the centre of the triangle. The ratio $|q/Q|$ so as to make the system in equilibrium is :-



(1) 1.3 (2) $1:\sqrt{3}$
(3) $\sqrt{3}:1$ (4) $2:\sqrt{3}$

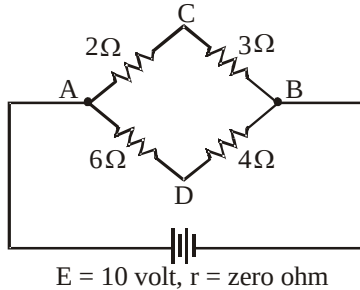
12. The equivalent resistance of the network shown in the figure between the points A and B is :-



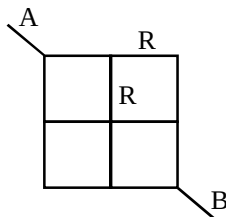
(1) 6Ω (2) 8Ω
(3) 16Ω (4) 24Ω

SPACE FOR ROUGH WORK

13. Four resistors 2Ω , 3Ω , 6Ω and 4Ω are connected to a battery of emf 4 volt and negligible internal resistance as shown in the following figure. Then, the potential difference between the points C and D is :-



- (1) -2 volt (2) $+2 \text{ volt}$
(3) $+1 \text{ volt}$ (4) -1 volt
14. A conductor wire having 10^{29} free electrons/ m^3 carries a current of 20 A. If the cross-section of the wire is 1mm^2 , then the drift velocity of electrons will be :- ($e = 1.6 \times 10^{-19} \text{ C}$)
- (1) $1.25 \times 10^{-5} \text{ ms}^{-1}$
(2) $6.25 \times 10^{-3} \text{ ms}^{-1}$
(3) $1.25 \times 10^{-4} \text{ ms}^{-1}$
(4) $1.25 \times 10^{-3} \text{ ms}^{-1}$
15. The equivalent resistance between the point A and B is (R is the resistance of each side of smaller square) :-



- (1) R (2) $\frac{3R}{2}$ (3) $2R$ (4) $R/2$

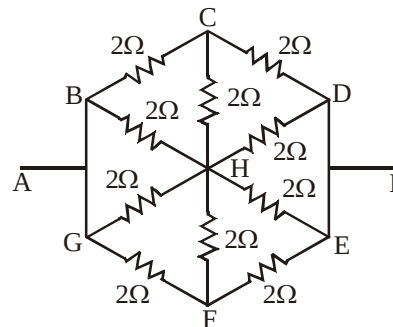
16. Mark the correct statement about the electric lines of force :-

- (1) The number of lines of force leaving a +ve charge or entering a -ve charge is proportional to the magnitude of charge
(2) The path of a charged particle in an electric field is not always along the field lines
(3) The direction of electric lines of force is along the normal to the conducting surfaces
(4) All of the above

17. The ratio of the resistance of conductor at temperature 15°C to its resistance at temperature 37.5°C is 4 : 5. the temperature coefficient of resistance of the conductor is :- (reference is taken as 0°C)

- (1) $\frac{1}{25}^\circ\text{C}^{-1}$ (2) $\frac{1}{50}^\circ\text{C}^{-1}$
(3) $\frac{1}{80}^\circ\text{C}^{-1}$ (4) $\frac{1}{75}^\circ\text{C}^{-1}$

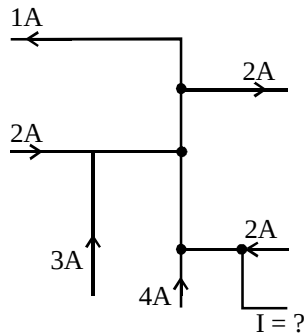
18. The effective resistance across the points A and I is :-



- (1) 2Ω (2) 1Ω
(3) 0.5Ω (4) 5Ω

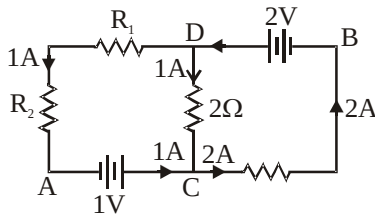
SPACE FOR ROUGH WORK

19. The magnitude and direction of current I (in A) indicated in the adjoining circuit is :-



- (1) $14 \rightarrow$ (2) $8 \rightarrow$
(3) $\leftarrow 4$ (4) $\leftarrow 8$

20. In the circuit shown below, if the potential at point A is taken to be zero, the potential at point B is :-



- (1) $+1V$
(2) $-1V$
(3) $+2V$
(4) $-2V$

21. Two equal negative charges each $-q$, are placed at points $(0, a)$ and $(0, -a)$ on y -axis. A positive charge q is released from point $(2a, 0)$. This charge will be :-
(1) execute SHM about the origin
(2) oscillate but not execute SHM
(3) move towards origin and will become stationary
(4) execute SHM along y -axis

22. A spherical conducting shell has a charge q . If a point charge Q is placed at the centre, then the charge on inner and outer surface of the shell respectively are :-

- (1) $0, Q$ (2) $-Q, Q + q$
(3) $q, Q - q$ (4) $Q, Q + q$

23. A parallel plate capacitor has capacitance of 10^{-12} F. The separation of the plates is doubled and wax is inserted between them which increases the capacitance to 2×10^{-12} F. The dielectric constant of wax is :

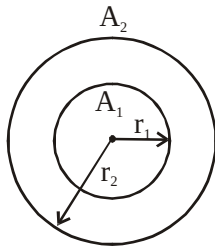
- (1) 2 (2) 3
(3) 4.0 (4) 8.0

24. Two cells of emf's E_1 and E_2 and internal resistance r_1 and r_2 are connected in parallel. Then the emf and internal resistance of the equivalent source are:

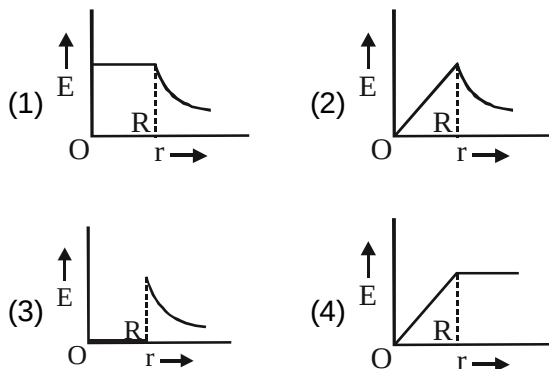
- (1) $(E_1 + E_2)$ and $\left(\frac{r_1 r_2}{r_1 + r_2} \right)$
(2) $(E_1 - E_2)$ and $(r_1 + r_2)$
(3) $\left(\frac{E_1 r_2 + E_2 r_1}{r_1 + r_2} \right)$ and $\left(\frac{r_1 r_2}{r_1 + r_2} \right)$
(4) $\left(\frac{E_1 r_2 + E_2 r_1}{r_1 + r_2} \right)$ and $(r_1 + r_2)$

SPACE FOR ROUGH WORK

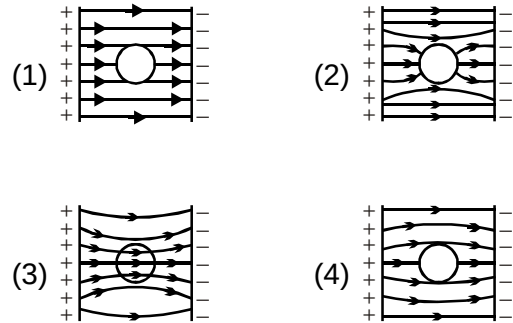
25. The capacity of a isolated parallel plate condenser is $5\mu\text{F}$. When a glass plate is placed between the plates of the capacitor, its potential becomes $1/8^{\text{th}}$ of the original value. The value of dielectric constant will be
(1) 1.6 (2) 5 (3) 8 (4) 40
26. Two spherical conductors A_1 and A_2 of radii r_1 and r_2 are placed concentrically in air. Then the equivalent capacitance of the system, if outer shell is earthed:



- (1) $\frac{4\pi\epsilon_0 r_1 r_2}{r_2 - r_1}$ (2) $4\pi\epsilon_0(r_2 + r_1)$
- (3) $\frac{4\pi\epsilon_0 K r_1 r_2}{r_2 - r_1}$ (4) $4\pi\epsilon_0 r_1$
27. The electric field due to a uniformly charged conducting sphere of radius R as a function of the distance from its centre is represented graphically by :-



28. An uncharged metal sphere is placed between two equal and oppositely charged metal plates. The nature of lines of force will be :-

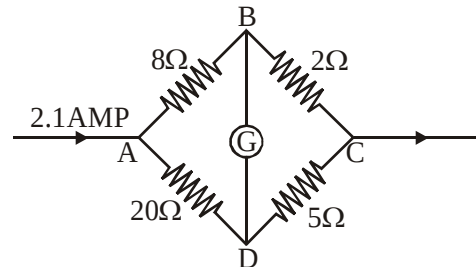


29. At any point on the perpendicular bisector of the line joining the two equal and opposite charges :-
(1) the electric field is zero
(2) the electric potential is zero
(3) the electric field is perpendicular to the line joining the two charges
(4) the electric potential decreases with the distance from the centre of the charges.
30. If E and V represent electric field and electric potential at a point respectively, then :-
(1) if $E = 0$, V may be zero or constant
(2) if $E = 0$, V must be zero
(3) if $V = 0$, E must be zero
(4) if $E = \text{constant}$, V must be zero

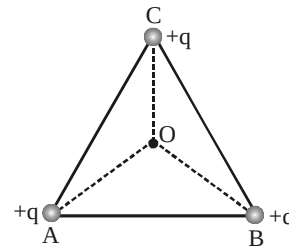
SPACE FOR ROUGH WORK

31. If an electron is thrown in the direction of electric field with some velocity then :-
 (1) its kinetic energy decreases but potential energy will increase
 (2) both kinetic and potential energy will increase
 (3) both kinetic and potential energy will decrease
 (4) kinetic energy will increase but potential energy will decrease
32. The drift velocity of free electrons in a conductor is v , when a current i is flowing in it. If both the radius and current are doubled, then the drift velocity will be :-
 (1) v
 (2) $v/2$
 (3) $v/4$
 (4) $v/8$
33. A parallel plate capacitor of capacitance C consists of two identical plates A and B. A charge q is given to plate A and charge $-q$ is given to plate B. The space between plates is vacuum. The separation between plates is d . The electric intensity at a point situated between plates is :
 (1) q/Cd
 (2) $q/2Cd$
 (3) $3q/2Cd$
 (4) none of these

34. In the given Fig. when galvanometer shows no deflection. Current flowing through $5\ \Omega$ resistance will be :



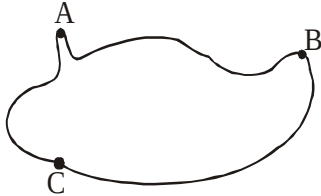
- (1) 0.5 A (2) 0.6 A
 (3) 1.5 A (4) 2.0 A
35. Like charges of equal magnitude q are kept at the vertices of an equilateral triangle ABC as shown in the fig. The electric intensity E and potential V , at the centroid O , are respectively :-



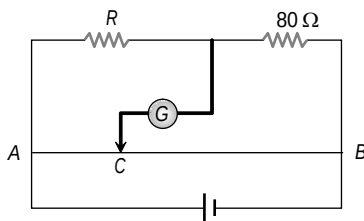
- (1) $E = 0, V = 0$
 (2) $E = 0, V \neq 0$
 (3) $E \neq 0, V = 0$
 (4) $E \neq 0, V \neq 0$

SPACE FOR ROUGH WORK

36. Figure shows a charged conductor of irregular shape. If σ_A , σ_B and σ_C are the surface charge densities at A, B and C respectively, then :-

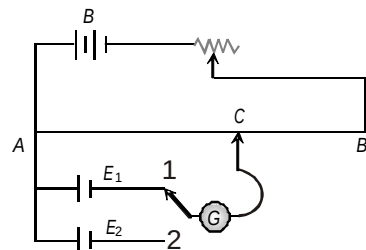


- (1) $\sigma_A = \sigma_B = \sigma_C$ (2) $\sigma_A < \sigma_B < \sigma_C$
(3) $\sigma_A > \sigma_B > \sigma_C$ (4) $\sigma_A = \sigma_B < \sigma_C$
37. The energy of a isolated charged capacitor is given by the expression by $U = \frac{q^2}{2C}$ (q = charge on the conductor and C = its capacity). If a dielectric slab of dielectric constant K is inserted between the plates, then its
- (1) Capacitance decreases
(2) electric field inside increases
(3) energy stored decreases
(4) None of these
38. AB is a wire of uniform resistance. The galvanometer G shows no current when the length $AC = 20\text{cm}$ and $CB = 80\text{ cm}$. The resistance R is equal to



- (1) 2Ω
(2) 8Ω
(3) 20Ω
(4) 40Ω

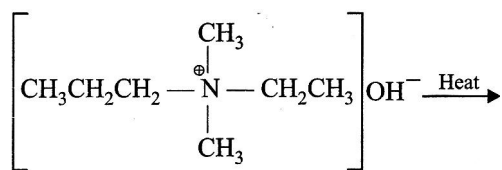
39. Choose the correct statements
- (1) electric field inside the current carrying wire is non-zero.
(2) electric field outside the current carrying wire is zero.
(3) Both are correct
(4) None of the above are correct
40. The circuit shown here is used to compare the e.m.f. of two cells E_1 and E_2 ($E_1 > E_2$). The null point is at C when the galvanometer is connected to 1. When the galvanometer is connected to 2, the null point will be



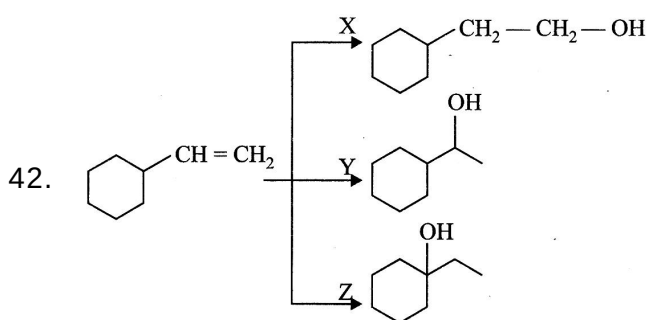
- (1) at the left of C
(2) at the right of C
(3) at C itself
(4) Nowhere on AB

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41. Consider the reaction
Which of the following is formed in major amount?

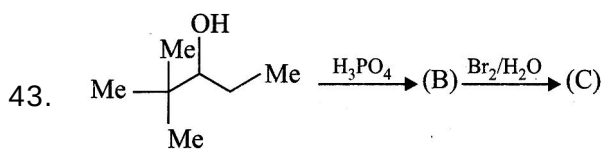


- (1) $\text{CH}_2 = \text{CH}_2$
 (2) $\text{CH}_3\text{CH} = \text{CH}_2$
 (3) Both (1) and (2) in equal amount
 (4) None, as no reaction takes place

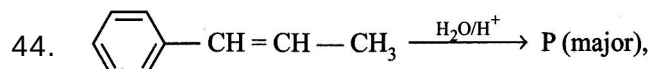
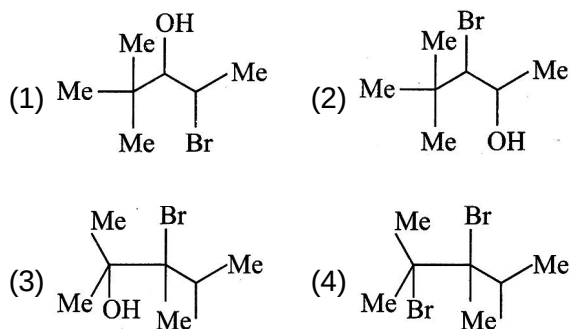


X, Y, Z reaction are :

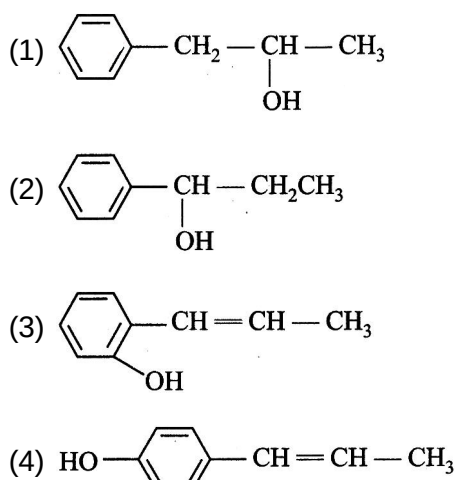
- (1) Simple hydration reaction
 (2) Hydroboration-oxidation, hydration and oxymercuration-demercuration
 (3) Hydroboration-oxidation, oxymercuration-demercuration and hydration
 (4) Oxymercuration-demercuration, hydroboration-oxidation and hydration



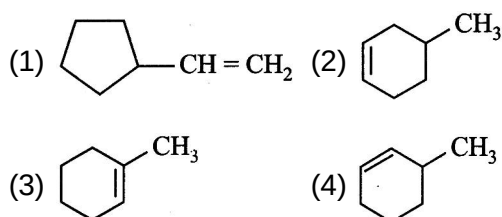
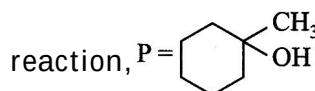
Compound (C) is :-



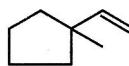
P is :-

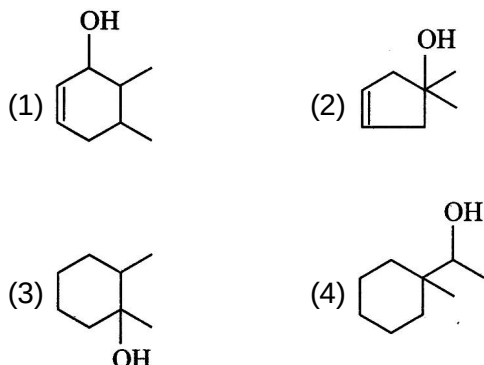


45. Which of the following alkene will give (P) on oxymercuration demercuration

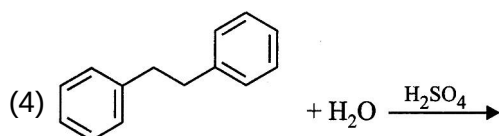
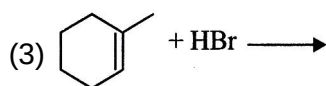
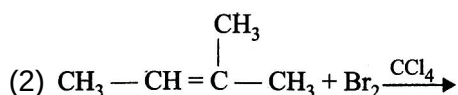
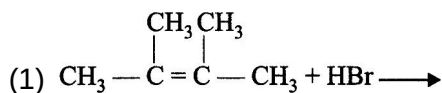


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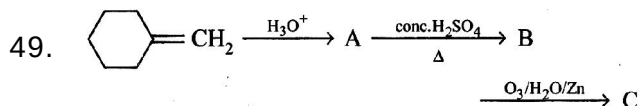
46.  P. Identify major product
'P' is:



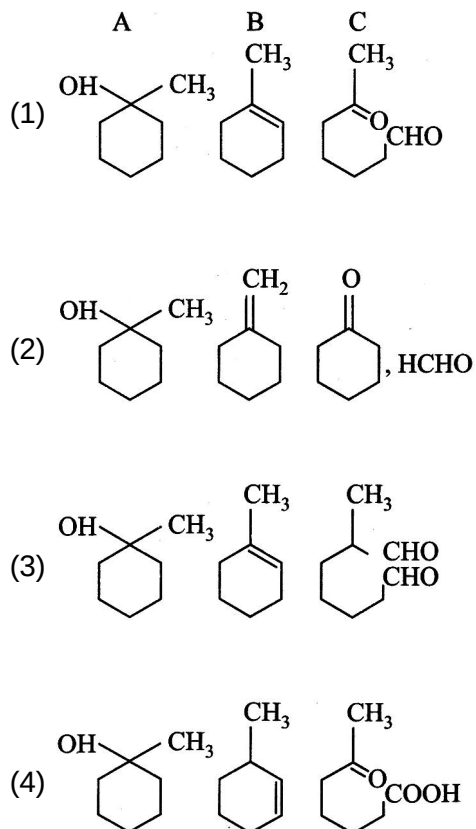
47. In which of the following reactions Markownikoff's rule of addition reaction is followed



48. Propene reacts with Cl_2 at 500°C the product formed is
(1) 1-chloro propene-1
(2) 2-chloro propene-1
(3) 1,2-dichloro propane
(4) 3-chloro propene-1

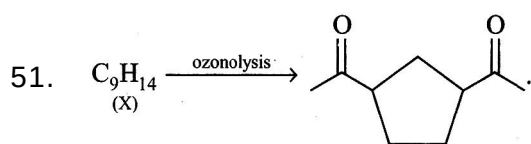


A, B and C are

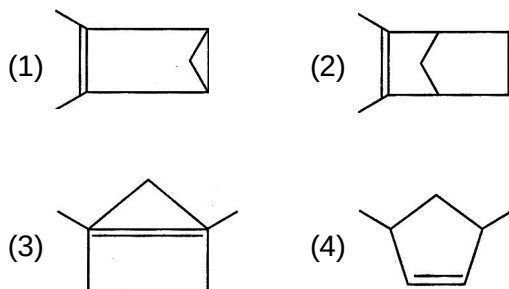


50. When HCl gas is passed through propene in the presence of benzoyl peroxide, it gives :-
(1) n-Propyl chloride
(2) 2-Chloropropane
(3) Allyl chloride
(4) No reaction

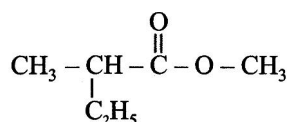
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Hence X is :-

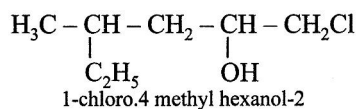


52. The correct IUPAC name of following compound is



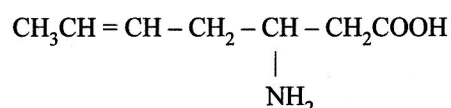
- (1) Methyl -2-ethylpropanoate
- (2) Methyl butane-2-carboxylate
- (3) Methyl- 2-methylbutanoate
- (4) Methoxypentanone

53. IUPAC name of



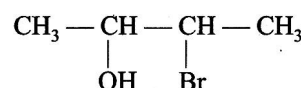
- (1) 1-chloro-4-methylhexan-2-ol
- (2) 1-chloro-4-methylhexan-2-al
- (3) 1-chloro-4-ethylpentan-2-ol
- (4) 1-chloro-2-hydroxy-4-methylhexane

54. The IUPAC name of the following is:



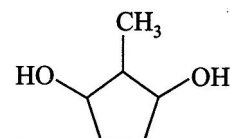
- (1) 3-Aminohept-5-enoic acid
- (2) 5-Aminohex-2-enecarboxylic acid
- (3) 3-Aminohept-4-enoic acid
- (4) 5-Aminohept-2-enoic acid

55. Total number of stereoisomers of compound is:



- (1) 2
- (2) 4
- (3) 6
- (4) 8

56. How many different stereoisomers exist for the compound below?



- (1) 2
- (2) 8
- (3) 6
- (4) 4

57. The number of isomers for the compounds with molecular formula $C_2BrClFI$ is:

- (1) 3
- (2) 4
- (3) 5
- (4) 6

58. How many cyclic isomers (structural and geometrical only) exist for C_5H_{10} ?

- (1) 3
- (2) 5
- (3) 6
- (4) 8

59. Which of the following statements concerning 3,4-dibromo-1-pentene and 3,5-dibromo-2-pentene are correct?

(I) They have same molecular formula $C_5H_8Br_2$

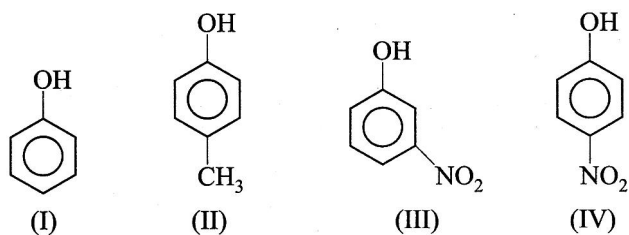
(II) They are positional isomers.

(III) They have similar chemical properties.

- (1) I and III
- (2) I and II
- (3) II and III
- (4) I, II and III

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60. In the following compounds,



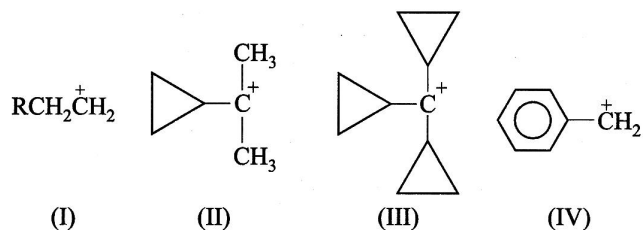
The order of acidity is :

- (1) III > IV > I > II (2) I > IV > III > II
(3) II > I > III > IV (4) IV > III > I > II

61. Which of the following is not an electrophile?

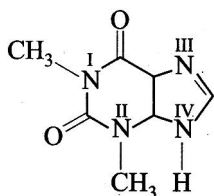
- (1) Cl^+ (2) $\text{CH}_2 = \text{CH}_2$
(3) Fe^{3+} (4) $^+\text{NO}_2$

62. In the following carbocations, the stability order is :-



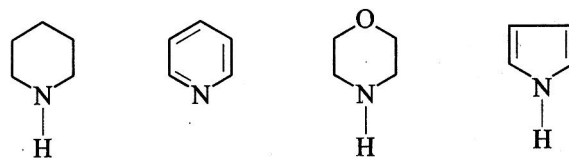
- (1) III > II > IV > I (2) IV > I > II > III
(3) IV > III > II > I (4) III > IV > II > I

63. The correct increasing order of basic strength of the labeled N is:



- (1) I < II < III < IV
(2) I < II < IV < III
(3) I < III < IV < II
(4) IV < III < II < I

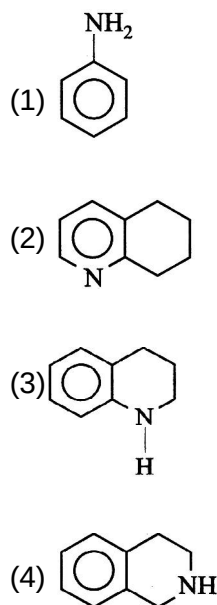
64. Select the basic strength order of following molecules.



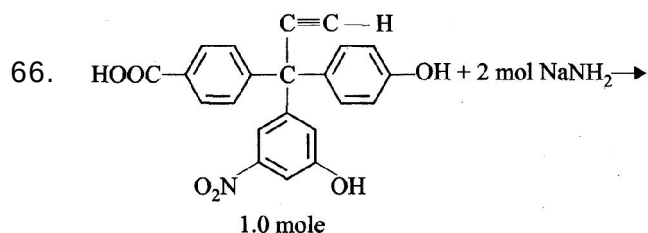
(I) Piperidine (II) Pyridine (III) Morpholine (IV) Pyrrole

- (1) (IV) > (I) > (III) > (II)
(2) (III) > (I) > (IV) > (II)
(3) (II) > (I) > (III) > (IV)
(4) (I) > (III) > (II) > (IV)

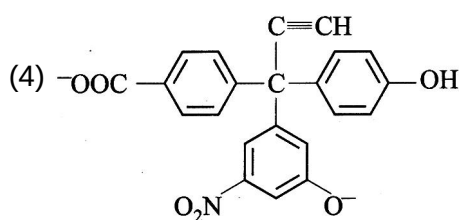
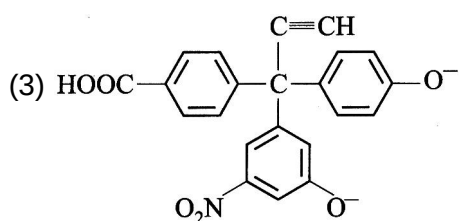
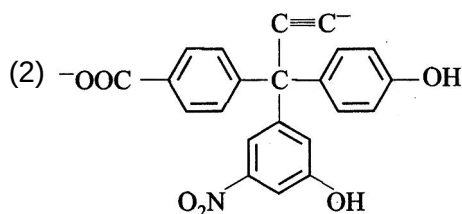
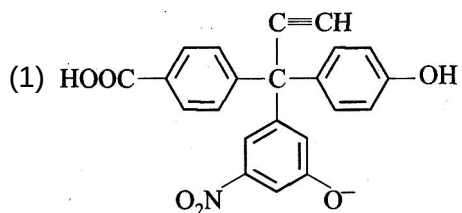
65. Choose the strongest base among the following :



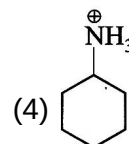
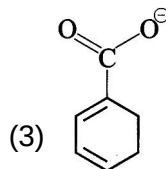
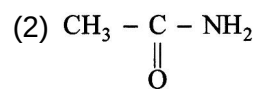
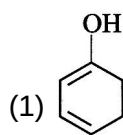
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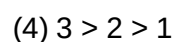
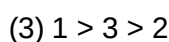
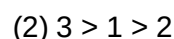
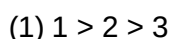
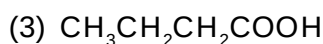
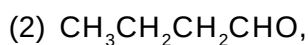
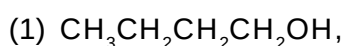
What is formed in the above reaction?



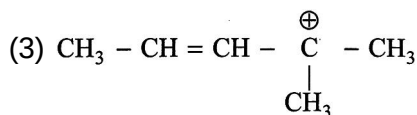
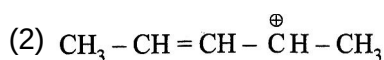
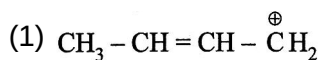
67. Which of the following cannot be a base?



68. Identify the correct order of boiling points of the following compounds:



69. Which allylic carbocation is most stable?



(4) All have same stability

70. Which of the following compounds is /are aromatic

(1) Benzene

(2) Phenol

(3) Benzoic Acid

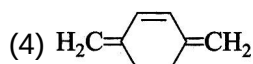
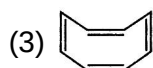
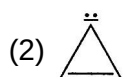
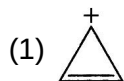
(4) All of these

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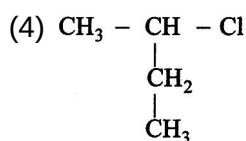
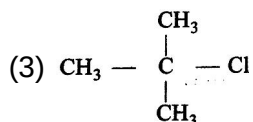
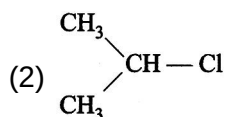
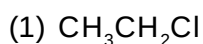
71. Number of π electrons present in naphthalene is

- (1) 4 (2) 6
(3) 10 (4) 14

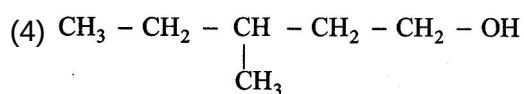
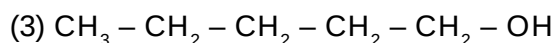
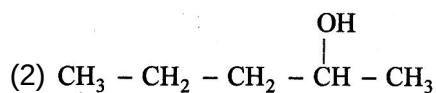
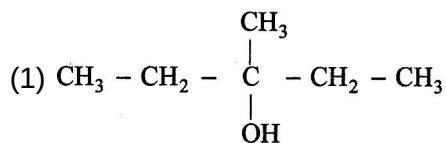
72. Which of the following compounds is aromatic in nature?



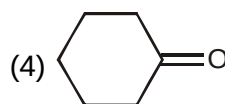
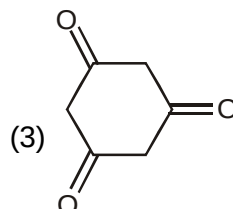
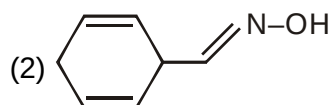
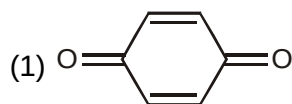
73. S_N1 reaction is fastest in



74. Among the following compounds which can be dehydrated very easily is

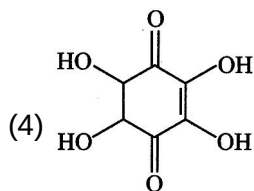
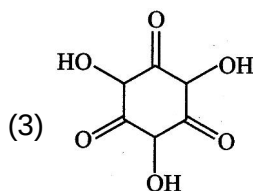
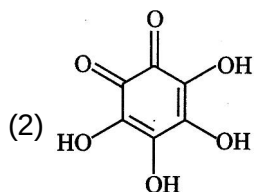
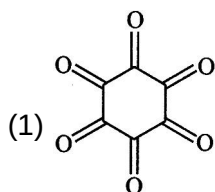
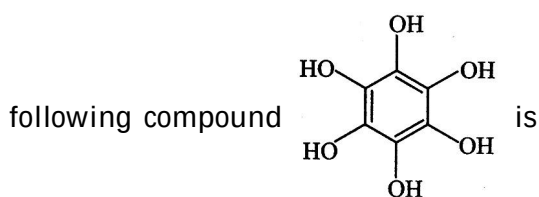


75. Tautomerism is not exhibited by :-

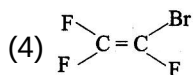
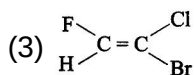
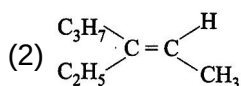
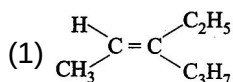


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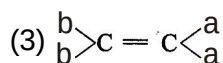
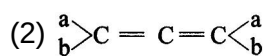
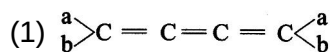
76. The most stable keto isomer of the



77. The Z-configuration in the following is

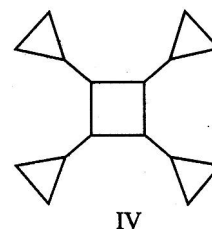
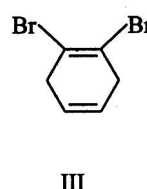
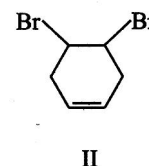
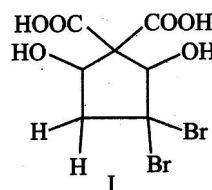


78. Which show geometrical isomerism?



(4) All of these

79. Which will show geometrical isomerism?



(1) I, II, III, IV

(2) I, II, III

(3) I, III, IV

(4) I, II, IV

80. The smallest alcohol which exhibit optical activity is

(1) n-Butyl alcohol

(2) Butan-1-ol

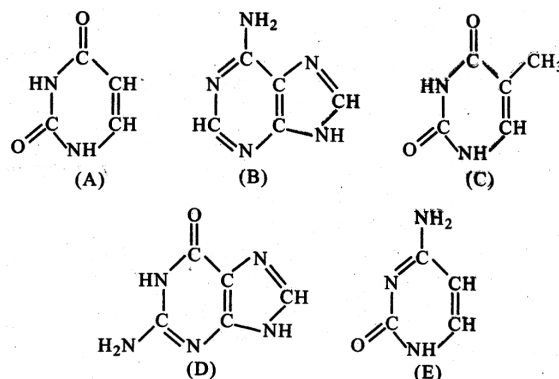
(3) Pentan-2-ol

(4) Butan-2-ol

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81. Cell division requires that the genetic material be able to :
- be replicated
 - undergo rare mutations
 - store information
 - all of these
82. Which of the following pairs is incorrectly matched?
- Linus Pauling — Isolated DNA for the first time
 - Rosalind franklin — Obtained x-ray diffraction images of DNA
 - Frederick Griffith— Discovered the phenomenon of transformation
 - Friedrich miescher — Discovered nucleic acid
83. A mixture of heat-killed NL-cells (dead cells) and live L-cells is injected into mice. NL-type cells are pathogenic, develop a disease and kill the animals. On the other hand L-type cells are nonpathogenic. The expected results could be :
- mice develop disease and die
 - 50% mice develop disease and die
 - mice die without developing disease
 - mice do not develop disease and do not die
84. Hershey and Chase demonstrated that DNA and not protein is the genetic material with the help of the radioactive isotopes:
- ^{59}Fe and ^{15}N
 - ^{32}S and ^{35}P
 - ^{15}N and ^{14}C
 - ^{35}S and ^{32}P
85. Prokaryotic DNA is:
- single stranded straight with protamines
 - single stranded circular without protamines
 - double stranded straight with histone
 - double stranded circular without histone
86. The sugar in DNA is:
- 2'-deoxyribose
 - 4'-deoxyribose
 - 3'-deoxyribose
 - 5'-deoxyribose

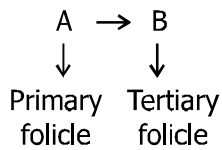
87. Read all the statements carefully.
- The four nucleotide bases are not necessarily present in DNA in exact equal proportions.
 - The total amount of purines are equal to the total amount of pyrimidines.
 - DNA ligase enzyme act to hydrolyse or breakdown a polynucleotide chain into its component nucleotides.
 - Nuclease enzymes are capable of restoring an intact DNA duplex.
- Of the above statements:
- A, B and C are correct but D is wrong
 - B, C and D are correct but A is wrong
 - B is correct but A, C and D are wrong
 - A and B are correct but C and D are wrong
88. The following diagrams represent the nitrogenous bases of nucleic acid molecules. Identify the correct combination :



- A = uracil, B = guanine, C = cytosine, D = adenine, E = thymine
 - A = thymine, B = adenine, C = cytosine, D = guanine, E = uracil
 - A = uracil, B = adenine, C = thymine, D = guanine, E = cytosine
 - A = uracil, B = guanine, C = thymine, D = adenine, E = cytosine
89. Which DNA molecule among the following will melt at lowest temperature?
- 5'-A-A-T-G-C-T-G-C-3'
3'-T-T-A-C-G-A-C-G-5'
 - 5'-A-A-T-A-A-A-G-C-3'
3'-T-T-A-T-T-T-C-G-5'
 - 5'-G-C-A-T-A-G-G-C-3'
3'-C-G-T-A-T-C-C-G-5'
 - 5'-A-T-G-G-C-T-G-C-3'
3'-T-A-C-C-G-A-C-G-5'

90. *Escherichia coli* fully labelled with ^{15}N is allowed to grow in ^{14}N medium. Compare density and resemblance of 1st generation DNA molecules w.r.t. parental DNA.
 (1) same density and resemble parent DNA.
 (2) different density but resemble parent DNA
 (3) same density but do not resemble parent DNA
 (4) different density and do not resemble parent DNA
91. DNA having labelled thymidine is allowed to replicate in medium having nonradioactive thymidine. After three duplications, the DNA molecules having labelled thymidine shall be :
 (1) one (2) two (3) four (4) eight
92. The anti-parallel nature of DNA refers to:
 (1) Its charged phosphate groups
 (2) The opposite direction of the two strands
 (3) The pairing of bases on one strand with bases on the other strand
 (4) The formation of hydrogen bonds between bases from opposite strands
93. Read the following statements and choose the correct option.
 A. Nitrogenous base is linked to the pentose sugar through a N-glycosidic linkage.
 B. Phosphate group is linked to 5'-OH of a nucleoside through phosphoester linkage.
 C. Two nucleosides are linked through 3' - 5' N-glycosidic linkage.
 D. Negatively charged DNA is wrapped around positively charged histone octamer to form nucleosome.
 E. The chromatin that is more densely packed and stains dark is called euchromatin.
 (1) A alone is wrong
 (2) D alone is wrong
 (3) C and E alone are wrong
 (4) A, B and D alone are wrong
94. Which statement about alleles is not true?
 (1) There may be several at a locus
 (2) One may be dominant over another
 (3) They may show incomplete dominance
 (4) They occupy different loci on the same chromosome
95. Hybrid vigour is mostly due to :
 (1) homozygosity of pure characters
 (2) superiority of all the genes
 (3) heterozygosity
 (4) All of the above
96. Which of the following characters selected by Mendel are recessive?
 (1) Yellow pod colour and wrinkled seeds
 (2) Green pod colour and dwarf plant
 (3) Dwarf plant and round seeds
 (4) Tall plant and axial flowers
97. Which statement about Mendel's cross of RR peas with rr peas is not true?
 (1) This is an example of a monohybrid cross
 (2) Three genotypes are observed in F_2 generation
 (3) Three phenotypes are observed in F_2 generation
 (4) Each parent can produce only one type of gamete
98. Which of the following crosses and resultant phenotypic ratios are mismatched?
- | Cross | Phenotypic ratio |
|------------------------|------------------|
| (1) $Tt \times Tt$ | — 3 : 1 |
| (2) $tt \times Tt$ | — 2 : 1 |
| (3) $TtYy \times ttyy$ | — 1 : 1 : 1 : 1 |
| (4) $TtYy \times TtYy$ | — 9 : 3 : 3 : 1 |
99. From a cross $AABb \times aaBb$, the genotypes $AaBB$, $AaBb$, $Aabb$, $aabb$ will be obtained in the following ratio :
 (1) 1 : 2 : 1 : 0 (2) 1 : 1 : 1 : 1
 (3) 0 : 3 : 1 : 0 (4) 1 : 1 : 1 : 0
100. The cytological event that corresponds to Mendel's law of independent assortment is seen in :
 (A) anaphase I (B) anaphase II
 (C) metaphase I (D) metaphase II
 (1) C, A (2) D, B
 (3) A, D (4) C, B

101. A linear diagrammatic representation of Oogenesis is given below



What are the name of stage A and B of life :

- (1) A = Birth, B= Adult reproductive life
- (2) A = Adult, B= Before birth
- (3) A = B= Adult reproductive life
- (4) A=2nd polar body , B=Ova.

102. Process of sperm formation is under the regulatary influence =

- (1) Oxytocin and vitamin A
- (2) FSH and vitamin E
- (3) LH and Vitamin E
- (4) Testosterone and vitamin C

103. How many sperms and ova would be produced from 175 primary spermetocyte and 60 secondary oocyte

- (1) 700 sperms and 240 ovum
- (2) 240 sperms and 700 ovum
- (3) 175 sperms and 240 ovum
- (4) 700 sperms and 60 ovum

104. How many sperms must have normal shape and size for normal fertility ?

- (1) At least 60 million / ejaculate
- (2) At least 40 million/ ejaculate
- (3) At least 100 million/ ejaculate
- (4) At least 180 million/ ejaculate

105. Select correct match

Column -I	Column-II
Part of sperm	Function
(P) Acrosome	(A) Genetic information
(Q) Nucleus	(B) Energy for the movement
(R) Middle part	(C) Enzymes that help in fertilisation

- (1) P –C, Q –A, R–B
- (2) P –A, Q –C, R–B
- (3) P –A, Q –B, R–C
- (4) P –B, Q –A, R–C

106. How many of the following hormones are produced by only placenta ?

- (I) hCG
- (II) hPL
- (III) Relaxin
- (IV) Prolactin
- (V) Thyroxin

- (1) 4
- (2) 3
- (3) 5
- (4) 2

107. During pregnancy the level of many hormones, like estrogens, progestogens, thyroxine etc, are increased several folds in maternal blood.

What is significance of this phenomenon ?

- (1) It is essential for supporting the fetal growth.
- (2) It is essential for maintenance of pregnancy.
- (3) It is essential for Oogenesis
- (4) Both (1) and (2)

108. In a normal pregnant woman, the amount of total gonadotropin activity was assessed.

The result expected was -

- (1) high level of circulating FSH and LH in the Uterus to stimulate implantation of the embryo.
- (2) High level of circulating hCG to stimulate endometrial thickening
- (3) High level of FSH and LH in uterus to stimulate endometrial thickening
- (4) High level of circulating hCG to stimulate oestrogen and progesterone synthesis.

109. Consider events given below -
A : Release of oxytocin from maternal pituitary.
B : Contractions in myometrium
Select correct statements :
(I) A is cause of B, during parturition
(II) B is cause of A, during parturition
(1) Only I (2) Only II
(3) Neither I nor II (4) Both I and II
110. Which hormone is known as somato mammatropic hormone ?
(1) hCG (2) hPL
(3) Oxytocin (4) Prolactin
111. Blastopore is found in -
(1) Blastula and is opening of blastocoel
(2) Blastula and opening of archenteron
(3) Gastrula and opening of blastocoel
(4) Gastrula and opening of archenteron
112. The best definition of the process of gastrulation is that, it is a process where the -
(1) Single layered blastula becomes two layered
(2) Zygote get converted in larva
(3) Archenteron is formed
(4) Cells move to occupy their definite position
113. How many statements of following are correct
(I) Uterine contractions which occur in the process of parturition are called as foetal ejection reflex.
(II) Abortions could happen spontaneously too.
(III) Placenta is expelled out of the uterus, soon after the infant is expelled
(1) 3 (2) 2
(3) 1 (4) 0
114. How many of the following statements with regard to embryonic development in human are correct -
(I) In morula amount of cytoplasm and DNA is equal to Zygote
(II) KI of blastomeres increases during cleavage
(III) There is no division of cytoplasm during cleavage
(IV) A three cell stage is found during cleavage
(1) 4 (2) 1 (3) 3 (4) 2
115. Select the correct function of Placenta
(1) Supply of oxygen to the embryo
(2) Supply of nutrients to the embryo
(3) Production of several hormones
(4) All
116. Which part of human embryo contains stem cells ?
(1) Trophoblast
(2) Ectoderm only
(3) Endoderm only
(4) Inner cell mass
117. How many eggs are released by a human ovary in the duration of the one year ?
(1) About 12 (2) About 6
(3) About 3 (4) About 1
118. During the development of embryo which of the following occurs first-
(1) Differentiation of tissue
(2) Differentiation of cell
(3) Differentiation of organs
(4) Differentiation of organ systems
119. Arrange A, B and C in correct order, according to size of individual cells
A = Zygote B = Morula C = Blastula
(1) A = B = C
(2) A > B > C
(3) A < B < C
(4) A = B < C
120. Cleavage starts when zygote moves towards :
(1) Ampulla to Isthmus
(2) Isthmus to Ampulla
(3) Infundibulum to Ampulla
(4) Ampulla to Infundibulum

Direction for Assertion & Reason questions.
 These questions consist of two statements each, printed as assertion and reason. while answering these questions you are required to choose any one of the following four responses.

- (1) If both Assertion & Reason are true & the reason is a correct explanation of the Assertion.
- (2) If both Assertion & Reason are true but reason is not a correct explanation of the Assertion.
- (3) If Assertion is true but the Reason is false
- (4) If both Assertion & Reason are false

121. *Assertion* : The electric bulbs glows immediately when switch is on.

Reason : The drift velocity of electrons in a metallic wire is very high.

122. *Assertion* : 1 C is practically a very big charge

Reason : The coulomb force is weaker than the gravitational force.

123. *Assertion* : A metallic shield in form of a hollow shell may be built to block an electric field.

Reason : In a hollow spherical shield, the electric field inside it is zero at every point

124. *Assertion* : Electrons move away from a low potential to high potential region.

Reason : Because electrons has negative charge

125. *Assertion* : If the distance between parallel plates of a air cored capacitor is halved and dielectric slab of dielectric constant 3 is inserted, then the capacitor becomes 6 times.

Reason : Capacity of the capacitor does not depend on the charge of capacitor

126. *Assertion* : Electric lines of force never cross each other.

Reason : Electric field follows super position principle.

127. *Assertion* : If a proton and an electron are placed in the same uniform electric field. They experience same acceleration.
Reason : Both electron & proton experience same force.

128. *Assertion* : When charges are shared between any two bodies, no charge is really lost, but some loss of energy does occur.

Reason : Energy is lost due to collision of electrons with atoms and other electrons inside conductor.

129. *Assertion* : The capacity of a given conductor remains same even if charge is varied on it.

Reason : Capacitance depends only shape and size of the capacitor and the medium side it

130. *Assertion* : Charge is invariant.

Reason : Charge does not depends on speed of frame of reference.

131. *Assertion* : Charge is quantized

Reason : Charge, which is less than 1 C is not possible

132. *Assertion* : On going away from a small electric dipole, electric field decreases.

Reason : Electric field varies inversely proportional to square of distance from an electric dipole.

133. *Assertion* : The connecting wires are made of copper.

Reason : The electrical conductivity of copper is high

134. *Assertion* : A potentiometer of longer length is used for accurate measurement.

Reason : Lesser the potential gradient, more is the sensitivity of potentiometer

135. *Assertion* : Electric field outside the conducting wire which carries a constant current is zero.

Reason : Net charge on conducting wire is zero.

136. *Assertion* : Bending a wire does not effect electrical resistance.

Reason : Resistance of wire is directly proportional to length of material.

137. *Assertion* : Electric potential energy of any positive charge is always positive.

Reason : Potential energy is a vector quantity.

138. *Assertion* : Two charged particles at rest experiences only electrical force

Reason : Charges at rest can only produce electric field.

139. *Assertion* : Current is a scalar quantity.

Reason : Electric current arises due to continuous flow of charged particles or ions.

140. *Assertion* : Electric field inside a metal surface is zero in electro static condition.

Reason : Excess Electric charge exise only on surface of conductor.

141. *Assertion* :  is aromatic.

Reason :  follows Huckel's rule.

142. *Assertion* : All the carbon atoms of but-2-ene lie in one plane.

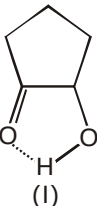
Reason : All the carbon atoms in but-2-ene are sp^2 -hybridized.

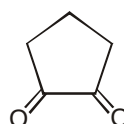
143. *Assertion* : Carbon-oxygen bonds are of equal length in acetate ion.

Reason : Bond length decreases with the multiplicity of bond between two atoms.

144. *Assertion* : Alkyl carbanions like ammonia have pyramidal shape.

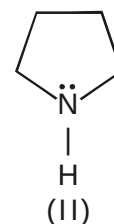
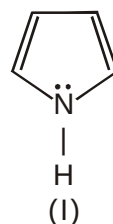
Reason : The carbon carrying negative charge has an octet of electrons.

145. *Assertion* :  is more stable than



Reason : 'I' has double bonds at anti position and hydrogen bonding while (II) has double bonds on the same side, so (II) has lot of electronic repulsion.

146. *Assertion* : Pyrrolidine (II) is more basic than pyrrole (I)



Reason : Protonated pyrrole has delocalisation of positive charge in aromatic ring.

147. *Assertion* : Tautomers differ both in the position of atoms as well as electrons.

Reason : CH_3CHO and $CH_2 = CHOH$ are resonance structure.

148. *Assertion* : Ortho substituted benzoic acids are stronger than benzoic acid.

Reason : Ortho substituent tends to prevent coplanarity of $-COOH$ with ring. Thus resonance is diminished which increases acidic strength (ortho effect).

149. *Assertion* : Free radicals are always planar.

Reason : They can achieve sp^2 as well as sp^3 -hybridisation.

150. *Assertion* : Addition of Br_2 to 1-butene two optical isomers.

Reason : The product contains one asymmetric carbon atom.

151. *Assertion* : Trans-2-butene on reaction with Br_2 gives meso-2, 3-dibromobutane.
Reason : The reaction involves syn-addition of bromine.
152. *Assertion* : Benzyl bromide when kept in acetone water produces benzyl alcohol.
Reason : The reaction follows $\text{S}_{\text{N}}2$ mechanism.
153. *Assertion* : $\text{Me}_3\overset{\cdot\cdot}{\text{C}}$ is more stable than $\overset{\cdot\cdot}{\text{C}}\text{H}_3$.
Reason : The +I effect of the three Me groups in $\text{Me}_3\overset{\cdot\cdot}{\text{C}}$ tends to make it more stable than $\overset{\cdot\cdot}{\text{C}}\text{H}_3$.
154. *Assertion* : Organic compounds which do not contain chiral carbon atoms cannot be optically active.
Reason : An organic compound is optically active only when its mirror image is non-superimposable.
155. *Assertion* : Stereoisomers which are not mirror images of each other are called diastereomers.
Reason : Diastereomers may not be optically active.
156. *Assertion* : Enol form of pentan-2, 4-dione is more stable than its keto-form.
Reason : It does not contain α -hydrogen atoms.
157. *Assertion* : Neopentane forms one mono substituted compound.
Reason : Neopentane is an isomer of pentane.
158. *Assertion* : Cyclohexanone exhibits keto-enol tautomerism.
Reason : In cyclohexanone, one form contains the keto group ($\text{C} = \text{O}$) while other contains enolic group ($-\text{C} = \text{C} - \text{OH}$).
159. *Assertion* : Staggered form is less stable than the eclipsed form.
Reason : The conformation in which the bond pairs of two central atoms are very near from one another is called staggered form.
160. *Assertion* : *Trans* isomers are more stable than *cis* isomer.
Reason : The *cis* isomer is the one in which two similar groups are on the same side of double bond.
161. *Assertion* : According to the law of independent assortment, characters segregate independently in F_2 generation.
Reason : Based on the observation of monohybrid cross, Mendel proposed second set of generalization that we call Mendel's Law of independent assortment.
162. *Assertion* : At F_2 stage in monohybrid cross, both parental traits are expressed in the proportion of 9 : 3.
Reason : The contrasting parental traits show blending at F_2 stage.
163. *Assertion* : Back cross is the cross between the F_1 progeny and either of the parent types.
Reason : Test cross is the cross between F_1 progeny and the double recessive genotype.
164. *Assertion* : DNA and RNA are the two types of nucleic acids present in living organisms.
Reason : DNA acts as a genetic material in most of the organisms.
165. *Assertion* : There is approximately a uniform distance between the two strands of DNA helix.
Reason : In DNA, always a purine comes opposite to a pyrimidine.
166. *Assertion* : If the sequence of bases in one strand is known then the sequence of the other strand can be predicted.
Reason : Both the strands of DNA are complementary to each other.

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| <p>167. <i>Assertion</i> : Nucleosomes in chromatin are observed as beads-on-string when viewed under an electron microscope.
<i>Reason</i> : A typical nucleosome contains 200 bp of DNA helix. Nucleosomes constitute the repeating units of structure in nucleus called chromatin, which are thread-like stained (coloured) bodies.</p> <p>168. <i>Assertion</i> : Both the polynucleotide chains of DNA have anti-parallel polarity.
<i>Reason</i> : If one chain has the polarity 5'→3', and the other has the polarity 3'→5'.</p> <p>169. <i>Assertion</i> : A nucleotide has three components –a nitrogen base, pentose sugar and a phosphate group.
<i>Reason</i> : Pentose sugar is a ribose in RNA and deoxyribose in DNA.</p> <p>170. <i>Assertion</i> : RNA acts as a genetic material in some viruses.
<i>Reason</i> : RNA mostly functions as a messenger.</p> <p>171. <i>Assertion</i> : Spermatogenesis starts at the age of puberty.
<i>Reason</i> : There is significant increase in the secretion of gonadotropins releasing hormone (GnRH) at puberty</p> <p>172. <i>Assertion</i> : The middle piece of sperm possess numerous mitochondria
<i>Reason</i> : Sperm motility is essential for fertilization</p> <p>173. <i>Assertion</i> : No more oogonia are formed or added after birth in female.
<i>Reason</i> : Oogenesis is initiated during the embryonic development stage of a female</p> | <p>174. <i>Assertion</i> : The secondary oocyte forms a new membrane called zona pellucida.
<i>Reason</i> : The cells of zona pellucida are disintegrated by the enzymes of acrosome of sperm</p> <p>175. <i>Assertion</i> : Menstruation can also occur if the released ovum is fertilized
<i>Reason</i> : Lack of menstruation is always indicative of pregnancy</p> <p>176. <i>Assertion</i> : The secretion of LH and FSH increases gradually during the follicular phase
<i>Reason</i> : Both LH and FSH attain a peak level in the middle of cycle</p> <p>177. <i>Assertion</i> : Mestrual cycle ceases around 50 year of age
<i>Reason</i> : All the ovarian follicles disintegrate at this age</p> <p>178. <i>Assertion</i> : All copulations do not lead to fertilisation.
<i>Reason</i> : Fertilisation can only occur if the ovum and sperms are transported simultaneously to the ampullary region</p> <p>179. <i>Assertion</i> : In morula stage, the cell divides without increases in size.
<i>Reason</i> : Zona pellucida remains till cleavage.</p> <p>180. <i>Assertion</i> : Zona pellucida help in ectopic pregnancy.
<i>Reason</i> : Zona pellucida persist even after fertilization.</p> |
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